P25 — Master Revision Sheet

298 Must-Solve Questions | 24 Interview Patterns | Use AFTER completing P01–P24

Revise by pattern, not by topic. Interviewers test patterns.

Goal: Recognise the pattern within 30 seconds of reading any problem.

Spend 2–3 weeks on this sheet before sitting for placement rounds.

PAT-01 — Two Pointers [14 Questions]

When to use: Sorted array + pair/triplet sum problems. In-place partition. Left and right converge.

#	Problem	Topic	Plat	Link	Key Insight
1	Two Sum II — Sorted Input	Arrays	LC	https://leetcode.com/problems/two-sum-ii-input-array-is-sorted/	Converge left + right
2	3Sum — All Zero Triplets	Arrays	LC	https://leetcode.com/problems/3sum/	Sort, fix one, two-ptr rest
3	3Sum Closest	Arrays	LC	https://leetcode.com/problems/3sum-closest/	Track min |diff| at each step
4	Container with Most Water	Arrays	LC	https://leetcode.com/problems/container-with-most-water/	Shrink smaller wall
5	Trapping Rain Water — Two Ptr	Arrays	LC	https://leetcode.com/problems/trapping-rain-water/	Left-max and right-max merge
6	Remove Duplicates from Sorted Array	Arrays	LC	https://leetcode.com/problems/remove-duplicates-from-sorted-array/	Write pointer skips repeats
7	Move Zeroes	Arrays	LC	https://leetcode.com/problems/move-zeroes/	Write pointer, skip zeroes
8	Valid Palindrome	Strings	LC	https://leetcode.com/problems/valid-palindrome/	Two ptrs skip non-alnum
9	Valid Palindrome II — One Deletion	Strings	LC	https://leetcode.com/problems/valid-palindrome-ii/	Try skipping left or right
10	Palindrome Linked List	LL	LC	https://leetcode.com/problems/palindrome-linked-list/	Find mid, reverse half
11	Linked List Cycle II — Entry Point	LL	LC	https://leetcode.com/problems/linked-list-cycle-ii/	Floyd then reset one to head
12	Middle of Linked List	LL	LC	https://leetcode.com/problems/middle-of-the-linked-list/	Fast moves 2x, slow moves 1x
13	Boats to Save People	Arrays	LC	https://leetcode.com/problems/boats-to-save-people/	Sort, heaviest + lightest if possible
14	4Sum	Arrays	LC	https://leetcode.com/problems/4sum/	Two fixed loops + two ptrs

Total PAT-01: 14 questions

PAT-02 — Sliding Window [14 Questions]

When to use: Subarray or substring problems with a contiguous constraint. Expand right, shrink left on violation.

#	Problem	Topic	Plat	Link	Key Insight
1	Longest Substring Without Repeating Chars	Strings	LC	https://leetcode.com/problems/longest-substring-without-repeating-characters/	HashMap last-seen + window
2	Minimum Window Substring	Strings	LC	https://leetcode.com/problems/minimum-window-substring/	Need-count, shrink when valid
3	Find All Anagrams in a String	Strings	LC	https://leetcode.com/problems/find-all-anagrams-in-a-string/	Fixed window, freq compare

#	Problem	Topic	Plat	Link	Key Insight
4	Longest Repeating Character Replacement	Strings	LC	https://leetcode.com/problems/longest-repeating-character-replacement/	maxFreq + (win-maxFreq) <= k
5	Permutation in String	Strings	LC	https://leetcode.com/problems/permutation-in-string/	Fixed window freq match
6	Max Consecutive Ones III	Arrays	LC	https://leetcode.com/problems/max-consecutive-ones-iii/	Count 0s in window, <= k
7	Fruit into Baskets (At Most 2 Types)	Arrays	LC	https://leetcode.com/problems/fruit-into-baskets/	HashMap distinct types <= 2
8	Sliding Window Maximum	Arrays	LC	https://leetcode.com/problems/sliding-window-maximum/	Monotonic deque of indices
9	Minimum Size Subarray Sum	Arrays	LC	https://leetcode.com/problems/minimum-size-subarray-sum/	Shrink left once sum >= target
10	Subarrays with K Different Integers	Arrays	LC	https://leetcode.com/problems/subarrays-with-k-different-integers/	atMost(k) - atMost(k-1)
11	Maximum Points from Cards	Arrays	LC	https://leetcode.com/problems/maximum-points-you-can-obtain-from-cards/	Window of size n-k in middle
12	Binary Subarrays with Sum K	Arrays	LC	https://leetcode.com/problems/binary-subarrays-with-sum/	atMost(k) - atMost(k-1)
13	Longest Subarray of 1s After Deleting One	Arrays	LC	https://leetcode.com/problems/longest-subarray-of-1s-after-deleting-one-element/	Zeros in window <= 1
14	Shortest Subarray with Sum >= K	Arrays	LC	https://leetcode.com/problems/shortest-subarray-with-sum-at-least-k/	Monotonic deque on prefix sums

Total PAT-02: 14 questions

PAT-03 — Binary Search Classic [14 Questions]

When to use: Eliminate half the search space each step. Use on sorted arrays or any monotone predicate.

#	Problem	Topic	Plat	Link	Key Insight
1	Binary Search	Arrays	LC	https://leetcode.com/problems/binary-search/	Core template: lo, hi, mid
2	Find First and Last Position	Arrays	LC	https://leetcode.com/problems/find-first-and-last-position-of-element-in-sorted-array/	Lower + upper bound
3	Search in Rotated Sorted Array	Arrays	LC	https://leetcode.com/problems/search-in-rotated-sorted-array/	Identify sorted half
4	Find Minimum in Rotated Sorted Array	Arrays	LC	https://leetcode.com/problems/find-minimum-in-rotated-sorted-array/	Min is at rotation pivot
5	Find Peak Element	Arrays	LC	https://leetcode.com/problems/find-peak-element/	Compare mid with mid+1
6	Single Element in Sorted Array	Arrays	LC	https://leetcode.com/problems/single-element-in-a-sorted-array/	Even/odd index check
7	Search in Rotated Array II (Duplicates)	Arrays	LC	https://leetcode.com/problems/search-in-rotated-sorted-array-ii/	Shrink both ends on equal
8	Median of Two Sorted Arrays	Arrays	LC	https://leetcode.com/problems/median-of-two-sorted-arrays/	Binary partition on shorter
9	Count Occurrences in Sorted Array	Arrays	GFG	https://practice.geeksforgeeks.org/problems/number-of-occurrence2259/	upperBound - lowerBound

#	Problem	Topic	Plat	Link	Key Insight
10	Find K Closest Elements	Arrays	LC	https://leetcode.com/problems/find-k-closest-elements/	BS for left boundary of window
11	Sqrt(x) — Integer	Math	LC	https://leetcode.com/problems/sqrtx/	BS on [0, x//2+1]
12	Search a 2D Matrix	Matrix	LC	https://leetcode.com/problems/search-a-2d-matrix/	Treat as 1D with index math
13	Time Based Key-Value Store	Design	LC	https://leetcode.com/problems/time-based-key-value-store/	BS on sorted timestamps list
14	H-Index II	Arrays	LC	https://leetcode.com/problems/h-index-ii/	BS: citations[mid] >= n-mid

Total PAT-03: 14 questions

PAT-04 — Binary Search on Answer [12 Questions]

When to use: Find minimum X such that condition(X) is true. BS on the answer space, check feasibility in O(n).

#	Problem	Topic	Plat	Link	Key Insight
1	Koko Eating Bananas	Arrays	LC	https://leetcode.com/problems/koko-eating-bananas/	Min speed: check hours at each speed
2	Capacity to Ship Packages in D Days	Arrays	LC	https://leetcode.com/problems/capacity-to-ship-packages-within-d-days/	Min capacity: count days
3	Split Array Largest Sum	Arrays	LC	https://leetcode.com/problems/split-array-largest-sum/	Min max: count splits
4	Minimum Days to Make M Bouquets	Arrays	LC	https://leetcode.com/problems/minimum-number-of-days-to-make-m-bouquets/	Count consecutive k-groups
5	Aggressive Cows	Arrays	GFG	https://practice.geeksforgeeks.org/problems/aggressive-cows/	Max min distance: place cows greedily
6	Book Allocation — Min Max Pages	Arrays	GFG	https://practice.geeksforgeeks.org/problems/allocate-minimum-number-of-pages0937/	Count students needed at each max
7	Painter's Partition	Arrays	GFG	https://practice.geeksforgeeks.org/problems/the-painters-partition-problem1535/	Count painters at each max length
8	Find Smallest Divisor Given Threshold	Math	LC	https://leetcode.com/problems/find-the-smallest-divisor-given-a-threshold/	Sum of ceil(n/d) <= threshold
9	Magnetic Force Between Two Balls	Arrays	LC	https://leetcode.com/problems/magnetic-force-between-two-balls/	Max min force: place balls greedily
10	Minimum Speed to Arrive on Time	Math	LC	https://leetcode.com/problems/minimum-speed-to-arrive-on-time/	Check if speed is feasible
11	Maximum Candies Allocated to K Children	Arrays	LC	https://leetcode.com/problems/maximum-candies-allocated-to-k-children/	Count piles of size m
12	Cutting Ribbons	Arrays	LC	https://leetcode.com/problems/cutting-ribbons/	Count ribbons of length L >= k

Total PAT-04: 12 questions

PAT-05 — Prefix Sum + HashMap [12 Questions]

When to use: Store prefix totals in a map to answer range sum / count queries in O(1). Core pattern for subarray problems.

#	Problem	Topic	Plat	Link	Key Insight
1	Subarray Sum Equals K	Arrays	LC	https://leetcode.com/problems/subarray-sum-equals-k/	Count prefix[j]-k in map
2	Contiguous Array (Equal 0s 1s)	Arrays	LC	https://leetcode.com/problems/contiguous-array/	Map -1 for 0, find zero-sum subarray
3	Subarray Sums Divisible by K	Arrays	LC	https://leetcode.com/problems/subarray-sums-divisible-by-k/	Prefix mod + count map
4	Longest Subarray Sum K (Negatives)	Arrays	GFG	https://practice.geeksforgeeks.org/problems/longest-sub-array-with-sum-k-1587115621/	First occurrence in prefix map
5	Path Sum III — Any Path in Tree	Trees	LC	https://leetcode.com/problems/path-sum-iii/	Prefix sum on root-to-node path
6	Count Subarrays with XOR = K	Arrays	GFG	https://practice.geeksforgeeks.org/problems/count-subarray-with-given-xor/	Prefix XOR, count XOR^k in map
7	Binary Subarrays with Sum	Arrays	LC	https://leetcode.com/problems/binary-subarrays-with-sum/	Prefix count array
8	Count Submatrices with Sum Target	Matrix	LC	https://leetcode.com/problems/number-of-submatrices-that-sum-to-target/	Compress to 1D + prefix sum map
9	Find Pivot Index	Arrays	LC	https://leetcode.com/problems/find-pivot-index/	Left sum = total - left - pivot
10	Product of Array Except Self	Arrays	LC	https://leetcode.com/problems/product-of-array-except-self/	Prefix product * suffix product
11	Count Subarrays with At Most K Odd Numbers	Arrays	LC	https://leetcode.com/problems/count-number-of-nice-subarrays/	Prefix odd count map
12	Minimum Consecutive Cards to Pick Up	Arrays	LC	https://leetcode.com/problems/minimum-consecutive-cards-to-pick-up/	Last occurrence HashMap

Total PAT-05: 12 questions

PAT-06 — Monotonic Stack [12 Questions]

When to use: Maintain a stack in increasing or decreasing order. Solves next-greater, histogram, and span problems in O(n).

#	Problem	Topic	Plat	Link	Key Insight
1	Daily Temperatures	Arrays	LC	https://leetcode.com/problems/daily-temperatures/	Decreasing stack of indices
2	Next Greater Element I	Arrays	LC	https://leetcode.com/problems/next-greater-element-i/	Stack + HashMap for lookup
3	Next Greater Element II — Circular	Arrays	LC	https://leetcode.com/problems/next-greater-element-ii/	Double the array traversal
4	Largest Rectangle in Histogram	Arrays	LC	https://leetcode.com/problems/largest-rectangle-in-histogram/	Left/right smaller boundaries
5	Maximal Rectangle — Matrix	Matrix	LC	https://leetcode.com/problems/maximal-rectangle/	Histogram row by row + stack
6	Sum of Subarray Minimums	Arrays	LC	https://leetcode.com/problems/sum-of-subarray-minimums/	Contribution = left_boundary * right_boundary
7	Sum of Subarray Ranges	Arrays	LC	https://leetcode.com/problems/sum-of-subarray-ranges/	Max contribution - min contribution
8	Trapping Rain Water — Stack	Arrays	LC	https://leetcode.com/problems/trapping-rain-water/	Pop when taller found, fill water
9	132 Pattern	Arrays	LC	https://leetcode.com/problems/132-pattern/	Track max third from right using stack

#	Problem	Topic	Plat	Link	Key Insight
10	Remove Duplicate Letters — Smallest	Strings	LC	https://leetcode.com/problems/remove-duplicate-letters/	Greedy stack with future-count
11	Remove K Digits — Smallest Number	Strings	LC	https://leetcode.com/problems/remove-k-digits/	Increasing stack, pop k times
12	Stock Span Problem	Arrays	GFG	https://practice.geeksforgeeks.org/problems/stock-span-problem-1587115621/	Stack of (price, span) pairs

Total PAT-06: 12 questions

PAT-07 — Tree — Traversal and Views [14 Questions]

When to use: Level order (BFS). DFS traversals. Views are first/last node per level. Vertical order groups by column.

#	Problem	Topic	Plat	Link	Key Insight
1	Binary Tree Level Order Traversal	Trees	LC	https://leetcode.com/problems/binary-tree-level-order-traversal/	Queue + level-size loop
2	Zigzag Level Order Traversal	Trees	LC	https://leetcode.com/problems/binary-tree-zigzag-level-order-traversal/	Alternate deque direction
3	Right Side View	Trees	LC	https://leetcode.com/problems/binary-tree-right-side-view/	Last node per BFS level
4	Left Side View	Trees	GFG	https://practice.geeksforgeeks.org/problems/left-view-of-binary-tree/	First node per BFS level
5	Top View of Binary Tree	Trees	GFG	https://practice.geeksforgeeks.org/problems/top-view-of-binary-tree/	BFS with column, first per column
6	Bottom View of Binary Tree	Trees	GFG	https://practice.geeksforgeeks.org/problems/bottom-view-of-binary-tree/	BFS with column, last per column
7	Vertical Order Traversal	Trees	LC	https://leetcode.com/problems/vertical-order-traversal-of-a-binary-tree/	DFS collect (col, row, val), sort
8	Average of Levels	Trees	LC	https://leetcode.com/problems/average-of-levels-in-binary-tree/	Sum / count per BFS level
9	Maximum Width of Binary Tree	Trees	LC	https://leetcode.com/problems/maximum-width-of-binary-tree/	Index nodes: 2i+1 and 2i+2
10	Binary Tree Inorder — Iterative	Trees	LC	https://leetcode.com/problems/binary-tree-inorder-traversal/	Stack: go left, process, go right
11	Boundary Traversal	Trees	GFG	https://practice.geeksforgeeks.org/problems/boundary-traversal-of-binary-tree/	Left spine + leaves + right spine
12	Spiral / Zigzag Traversal	Trees	GFG	https://practice.geeksforgeeks.org/problems/level-order-traversal-in-spiral-form/	Two stacks alternating
13	Cousins in Binary Tree	Trees	LC	https://leetcode.com/problems/cousins-in-binary-tree/	BFS: same level, different parent
14	All Nodes Distance K	Trees	LC	https://leetcode.com/problems/all-nodes-distance-k-in-binary-tree/	Parent map + BFS from target

Total PAT-07: 14 questions

PAT-08 — Tree — Advanced (Paths, LCA, Construction) [14 Questions]

When to use: Post-order for path sums and LCA. HashMap inorder index for O(n) construction.

#	Problem	Topic	Plat	Link	Key Insight
1	Diameter of Binary Tree	Trees	LC	https://leetcode.com/problems/diameter-of-binary-tree/	Post-order: left+right depth
2	Binary Tree Maximum Path Sum	Trees	LC	https://leetcode.com/problems/binary-tree-maximum-path-sum/	Max gain = max(0, left, right) + node
3	Lowest Common Ancestor — General	Trees	LC	https://leetcode.com/problems/lowest-common-ancestor-of-a-binary-tree/	Post-order return both targets up
4	Balanced Binary Tree	Trees	LC	https://leetcode.com/problems/balanced-binary-tree/	Post-order return -1 if unbalanced
5	Construct from Preorder + Inorder	Trees	LC	https://leetcode.com/problems/construct-binary-tree-from-preorder-and-inorder-traversal/	HashMap inorder index
6	Serialize and Deserialize Binary Tree	Trees	LC	https://leetcode.com/problems/serialize-and-deserialize-binary-tree/	Preorder with null markers
7	Path Sum II — All Paths	Trees	LC	https://leetcode.com/problems/path-sum-ii/	DFS backtrack with running sum
8	Flatten Binary Tree to Linked List	Trees	LC	https://leetcode.com/problems/flatten-binary-tree-to-linked-list/	Morris: thread right pointer
9	Sum Root to Leaf Numbers	Trees	LC	https://leetcode.com/problems/sum-root-to-leaf-numbers/	DFS: pass running number down
10	Burning Tree — Time to Burn	Trees	GFG	https://practice.geeksforgeeks.org/problems/burning-tree/	Parent map + BFS from node
11	Count Nodes Complete Tree — O(log^2 n)	Trees	LC	https://leetcode.com/problems/count-complete-tree-nodes/	Compare left/right heights
12	Binary Tree Cameras	Trees	LC	https://leetcode.com/problems/binary-tree-cameras/	Greedy: 3 states (uncovered/covered/camera)
13	House Robber III — Tree DP	Trees	LC	https://leetcode.com/problems/house-robber-iii/	Post-order: rob vs skip per node
14	Unique BSTs Count — Catalan	Trees	LC	https://leetcode.com/problems/unique-binary-search-trees/	dp[n] = sum dp[i]*dp[n-1-i]

Total PAT-08: 14 questions

PAT-09 — BST Patterns [12 Questions]

When to use: Exploit BST sorted property: inorder gives sorted array free, LCA is directional, range prunes branches.

#	Problem	Topic	Plat	Link	Key Insight
1	Validate BST	BST	LC	https://leetcode.com/problems/validate-binary-search-tree/	Pass valid (min, max) range down
2	Kth Smallest in BST	BST	LC	https://leetcode.com/problems/kth-smallest-element-in-a-bst/	Inorder with counter
3	LCA in BST	BST	LC	https://leetcode.com/problems/lowest-common-ancestor-of-a-binary-search-tree/	Split direction at each node
4	Convert BST to Greater Sum Tree	BST	LC	https://leetcode.com/problems/convert-bst-to-greater-tree/	Reverse inorder running sum
5	Delete Node in BST	BST	LC	https://leetcode.com/problems/delete-node-in-a-bst/	3 cases: leaf / 1 child / inorder succ
6	BST Iterator — O(h) Space	BST	LC	https://leetcode.com/problems/binary-search-tree-iterator/	Push all left spine to stack

#	Problem	Topic	Plat	Link	Key Insight
7	Recover BST — Two Swapped Nodes	BST	LC	https://leetcode.com/problems/recover-binary-search-tree/	Morris inorder, find 2 violations
8	Two Sum in BST	BST	LC	https://leetcode.com/problems/two-sum-iv-input-is-a-bst/	Inorder array + two pointer
9	Floor and Ceiling in BST	BST	GFG	https://practice.geeksforgeeks.org/problems/floor-in-bst/	Track best candidate while traversing
10	Construct BST from Preorder	BST	LC	https://leetcode.com/problems/construct-binary-search-tree-from-preorder-traversal/	Range-based recursion
11	Inorder Successor in BST	BST	GFG	https://practice.geeksforgeeks.org/problems/inorder-successor-in-bst/	Go right child or left-turn ancestor
12	Maximum Sum BST in Tree	BST	LC	https://leetcode.com/problems/maximum-sum-bst-in-binary-tree/	Post-order validate + track sum

Total PAT-09: 12 questions

PAT-10 — Graph — BFS [14 Questions]

When to use: BFS gives shortest path in unweighted graphs. Multi-source BFS from all starting nodes simultaneously.

#	Problem	Topic	Plat	Link	Key Insight
1	Word Ladder — Minimum Steps	Graphs	LC	https://leetcode.com/problems/word-ladder/	BFS on word graph, 1-char changes
2	Rotting Oranges — Multi-Source BFS	Graphs	LC	https://leetcode.com/problems/rotting-oranges/	Add all rotten, BFS level by level
3	01 Matrix — Distance to Nearest 0	Graphs	LC	https://leetcode.com/problems/01-matrix/	Multi-source from all 0s
4	Shortest Path in Binary Matrix	Graphs	LC	https://leetcode.com/problems/shortest-path-in-binary-matrix/	BFS 8-directional
5	Pacific Atlantic Water Flow	Graphs	LC	https://leetcode.com/problems/pacific-atlantic-water-flow/	Two BFS from borders, intersection
6	Walls and Gates — Fill Distances	Graphs	GFG	https://practice.geeksforgeeks.org/problems/walls-and-gates/	Multi-source BFS from gates
7	Open the Lock	Graphs	LC	https://leetcode.com/problems/open-the-lock/	BFS on state space (4-digit string)
8	Minimum Knight Moves	Graphs	LC	https://leetcode.com/problems/minimum-knight-moves/	BFS 8 knight moves
9	Snakes and Ladders	Graphs	LC	https://leetcode.com/problems/snakes-and-ladders/	BFS on board cells
10	Jump Game III — Can Reach Zero?	Graphs	LC	https://leetcode.com/problems/jump-game-iii/	BFS: index +nums[i] and -nums[i]
11	Time to Inform All Employees	Graphs	LC	https://leetcode.com/problems/time-needed-to-inform-all-employees/	BFS from CEO on tree
12	Minimum Height Trees	Graphs	LC	https://leetcode.com/problems/minimum-height-trees/	Prune leaves iteratively (BFS)
13	Bus Routes — Minimum Buses	Graphs	LC	https://leetcode.com/problems/bus-routes/	BFS on routes not stops
14	Shortest Path in Grid Eliminating Obstacles	Graphs	LC	https://leetcode.com/problems/shortest-path-in-a-grid-with-obstacles-elimination/	BFS state (r, c, k remaining)

Total PAT-10: 14 questions

PAT-11 — Graph — DFS [12 Questions]

When to use: DFS for connected components, cycle detection, coloring, and recursive grid exploration.

#	Problem	Topic	Plat	Link	Key Insight
1	Number of Islands	Graphs	LC	https://leetcode.com/problems/number-of-islands/	DFS mark visited
2	Number of Connected Components	Graphs	LC	https://leetcode.com/problems/number-of-connected-components-in-an-undirected-graph/	DFS/BFS count calls from unvisited
3	Max Area of Island	Graphs	LC	https://leetcode.com/problems/max-area-of-island/	DFS returns size of component
4	Surrounded Regions	Graphs	LC	https://leetcode.com/problems/surrounded-regions/	DFS from border Os, flip rest
5	Clone Graph	Graphs	LC	https://leetcode.com/problems/clone-graph/	DFS with HashMap original→clone
6	Detect Cycle Directed — DFS Colors	Graphs	GFG	https://practice.geeksforgeeks.org/problems/detect-cycle-in-a-directed-graph/	0=white 1=grey 2=black, grey→grey=cycle
7	Detect Cycle Undirected — DFS Parent	Graphs	GFG	https://practice.geeksforgeeks.org/problems/detect-cycle-in-an-undirected-graph/	Visited neighbour != parent
8	Keys and Rooms	Graphs	LC	https://leetcode.com/problems/keys-and-rooms/	DFS with keys as edges
9	Is Graph Bipartite — 2-Coloring	Graphs	LC	https://leetcode.com/problems/is-graph-bipartite/	DFS color, conflict = not bipartite
10	Critical Connections — Bridges	Graphs	LC	https://leetcode.com/problems/critical-connections-in-a-network/	Tarjan: low[v] > disc[u] = bridge
11	Redundant Connection II — Directed	Graphs	LC	https://leetcode.com/problems/redundant-connection-ii/	3 cases: two parents / cycle alone
12	Count Strongly Connected Components	Graphs	GFG	https://practice.geeksforgeeks.org/problems/strongly-connected-components-kosarajus-algorithm/	Kosaraju: DFS order + transpose DFS

Total PAT-11: 12 questions

PAT-12 — Shortest Path [12 Questions]

When to use: Dijkstra for non-negative weights. Bellman-Ford for negative weights. Floyd-Warshall for all pairs.

#	Problem	Topic	Plat	Link	Key Insight
1	Dijkstra's Algorithm	Graphs	GFG	https://practice.geeksforgeeks.org/problems/implementing-dijkstra-set-1-adjacency-matrix/	Min-heap (dist, node), relax neighbours
2	Network Delay Time	Graphs	LC	https://leetcode.com/problems/network-delay-time/	Dijkstra from source, max dist
3	Cheapest Flights K Stops — Bellman-Ford	Graphs	LC	https://leetcode.com/problems/cheapest-flights-within-k-stops/	Relax exactly K+1 times
4	Path with Minimum Effort	Graphs	LC	https://leetcode.com/problems/path-with-minimum-effort/	Dijkstra: minimize max edge on path
5	Swim in Rising Water	Graphs	LC	https://leetcode.com/problems/swim-in-rising-water/	Dijkstra or binary search + BFS
6	Minimum Cost Path in Grid	Graphs	LC	https://leetcode.com/problems/minimum-cost-to-make-at-least-one-valid-path-in-a-grid/	0-1 BFS deque

#	Problem	Topic	Plat	Link	Key Insight
7	Floyd-Warshall — All Pairs	Graphs	GFG	https://practice.geeksforgeeks.org/problems/implementing-floyd-warshall2118/	dp[i][j] via each intermediate k
8	Shortest Path Alternating Colors	Graphs	LC	https://leetcode.com/problems/shortest-path-with-alternating-colors/	BFS with edge color state
9	Number of Ways to Arrive at Destination	Graphs	LC	https://leetcode.com/problems/number-of-ways-to-arrive-at-destination/	Dijkstra + count array
10	Find City with Fewest Reachable in Threshold	Graphs	LC	https://leetcode.com/problems/find-the-city-with-the-smallest-number-of-neighbors-at-a-threshold-distance/	Floyd-Warshall + count
11	Minimum Fuel Cost to Report	Graphs	LC	https://leetcode.com/problems/minimum-fuel-cost-to-report-to-the-capital/	DFS: ceil(people/seats) cost per edge
12	Second Shortest Path	Graphs	LC	https://leetcode.com/problems/second-minimum-time-to-reach-destination/	BFS tracking 1st and 2nd distance

Total PAT-12: 12 questions

PAT-13 — Topological Sort [10 Questions]

When to use: Order DAG nodes so all edges point forward. Kahn's BFS or DFS with stack. Cycle = impossible order.

#	Problem	Topic	Plat	Link	Key Insight
1	Course Schedule I — Is Possible?	Graphs	LC	https://leetcode.com/problems/course-schedule/	Kahn's: cycle = not possible
2	Course Schedule II — The Order	Graphs	LC	https://leetcode.com/problems/course-schedule-ii/	Kahn's: return topological order
3	Alien Dictionary — Char Order	Graphs	LC	https://leetcode.com/problems/alien-dictionary/	Build char DAG, topo sort
4	Largest Color Value in DAG	Graphs	LC	https://leetcode.com/problems/largest-color-value-in-a-directed-graph/	Topo sort + DP: max color count
5	Parallel Courses — Max Levels	Graphs	LC	https://leetcode.com/problems/parallel-courses/	BFS topo: max BFS depth
6	Find Eventual Safe States	Graphs	LC	https://leetcode.com/problems/find-eventual-safe-states/	Reverse graph: Kahn's on reversed
7	Sort Items by Groups — Two-Level Topo	Graphs	LC	https://leetcode.com/problems/sort-items-by-groups-respecting-dependencies/	Topo on items + topo on groups
8	Minimum Time to Complete Tasks — Prereqs	Graphs	LC	https://leetcode.com/problems/parallel-courses-ii/	Topo + DP on DAG
9	Sequence Reconstruction — Unique Order?	Graphs	LC	https://leetcode.com/problems/sequence-reconstruction/	Topo: exactly one node with 0 in-degree at each step
10	Build Order — Dependency Resolution	Graphs	GFG	https://practice.geeksforgeeks.org/problems/topological-sort/	Kahn's or DFS stack

Total PAT-13: 10 questions

PAT-14 — 1D Dynamic Programming [18 Questions]

When to use: dp[i] depends on a few previous states. Define subproblem, write recurrence, handle base cases.

#	Problem	Topic	Plat	Link	Key Insight
1	Climbing Stairs	DP	LC	https://leetcode.com/problems/climbing-stairs/	dp[i] = dp[i-1] + dp[i-2]
2	House Robber	DP	LC	https://leetcode.com/problems/house-robber/	dp[i] = max(dp[i-1], dp[i-2]+nums[i])
3	House Robber II — Circular	DP	LC	https://leetcode.com/problems/house-robber-ii/	Two separate runs on sub-arrays
4	Coin Change — Minimum Coins	DP	LC	https://leetcode.com/problems/coin-change/	dp[amt] = min(dp[amt-c]+1)
5	Coin Change II — Count Ways	DP	LC	https://leetcode.com/problems/coin-change-2/	dp[amt] += dp[amt-c]
6	Decode Ways	DP	LC	https://leetcode.com/problems/decode-ways/	1-digit + 2-digit valid transitions
7	Word Break — Can Segment?	DP	LC	https://leetcode.com/problems/word-break/	dp[i] = any dp[j] + dict.contains(j..i)
8	Jump Game I	DP/Greedy	LC	https://leetcode.com/problems/jump-game/	Track max reachable
9	Jump Game II — Min Jumps	Greedy	LC	https://leetcode.com/problems/jump-game-ii/	BFS layers: update farthest reach
10	Maximum Subarray — Kadane	DP	LC	https://leetcode.com/problems/maximum-subarray/	dp[i] = max(nums[i], dp[i-1]+nums[i])
11	Maximum Product Subarray	DP	LC	https://leetcode.com/problems/maximum-product-subarray/	Track max and min (flip on negative)
12	Minimum Cost For Tickets	DP	LC	https://leetcode.com/problems/minimum-cost-for-tickets/	dp[i] = min of 3 pass options
13	Perfect Squares — Min Count	DP	LC	https://leetcode.com/problems/perfect-squares/	dp[i] = min(dp[i-sq]+1)
14	Delete and Earn	DP	LC	https://leetcode.com/problems/delete-and-earn/	Convert to House Robber variant
15	Best Time Buy/Sell with Cooldown	DP	LC	https://leetcode.com/problems/best-time-to-buy-and-sell-stock-with-cooldown/	State machine: hold / sold / rest
16	Best Time Buy/Sell with Fee	DP	LC	https://leetcode.com/problems/best-time-to-buy-and-sell-stock-with-transaction-fee/	dp: hold vs cash state
17	Fibonacci — Space Optimised	DP	LC	https://leetcode.com/problems/fibonacci-number/	Rolling two variables
18	Count Palindromic Substrings	DP	LC	https://leetcode.com/problems/palindromic-substrings/	Expand from each center

Total PAT-14: 18 questions

PAT-15 — 2D Grid DP [12 Questions]

When to use: dp[i][j] built from dp[i-1][j] and dp[i][j-1]. Grid traversal with overlapping subproblems.

#	Problem	Topic	Plat	Link	Key Insight
1	Unique Paths	DP	LC	https://leetcode.com/problems/unique-paths/	dp[i][j] = dp[i-1][j] + dp[i][j-1]
2	Unique Paths II — Obstacles	DP	LC	https://leetcode.com/problems/unique-paths-ii/	dp[i][j] = 0 if obstacle
3	Minimum Path Sum	DP	LC	https://leetcode.com/problems/minimum-path-sum/	dp[i][j] = grid + min(top, left)

#	Problem	Topic	Plat	Link	Key Insight
4	Maximal Square of 1s	DP	LC	https://leetcode.com/problems/maximal-square/	dp[i][j] = min(top, left, top-left) + 1
5	Triangle — Min Falling Path	DP	LC	https://leetcode.com/problems/triangle/	Bottom-up: dp[j] = min(dp[j], dp[j+1])
6	Dungeon Game	DP	LC	https://leetcode.com/problems/dungeon-game/	Reverse DP: min health to exit
7	Minimum Falling Path Sum	DP	LC	https://leetcode.com/problems/minimum-falling-path-sum/	dp[i][j] = grid + min 3 above
8	Count Square Submatrices All 1s	DP	LC	https://leetcode.com/problems/count-square-submatrices-with-all-ones/	dp[i][j] counts squares ending here
9	Number of Submatrices that Sum to Target	DP	LC	https://leetcode.com/problems/number-of-submatrices-that-sum-to-target/	Compress columns + prefix sum map
10	Paint House III — K Colors M Neighborhoods	DP	LC	https://leetcode.com/problems/paint-house-iii/	3D DP: house, color, neighborhood
11	Cherry Pickup II — Two Robots	DP	LC	https://leetcode.com/problems/cherry-pickup-ii/	3D DP: step, r1, r2
12	Longest Increasing Path in Matrix	DFS+Memo	LC	https://leetcode.com/problems/longest-increasing-path-in-a-matrix/	Memoised DFS on directed graph

Total PAT-15: 12 questions

PAT-16 — LCS / LIS Family [14 Questions]

When to use: LCS: longest match between two sequences. LIS: longest strictly increasing run. Many variants reduce here.

#	Problem	Topic	Plat	Link	Key Insight
1	Longest Common Subsequence	DP	GFG	https://practice.geeksforgeeks.org/problems/longest-common-subsequence-1587115620/	dp[i][j] = match: +1, else max skip
2	Longest Common Substring	DP	GFG	https://practice.geeksforgeeks.org/problems/longest-common-substring1458/	dp[i][j] = dp[i-1][j-1]+1 if match, else 0
3	Edit Distance	DP	LC	https://leetcode.com/problems/edit-distance/	min(insert, delete, replace)
4	Shortest Common Supersequence	DP	LC	https://leetcode.com/problems/shortest-common-supersequence/	LCS + remaining chars of both
5	Longest Palindromic Subsequence	DP	LC	https://leetcode.com/problems/longest-palindromic-subsequence/	LCS(s, reverse(s))
6	Minimum Insertions for Palindrome	DP	GFG	https://practice.geeksforgeeks.org/problems/form-a-palindrome1455/	n - LPS(s)
7	Longest Increasing Subsequence	DP	LC	https://leetcode.com/problems/longest-increasing-subsequence/	Patience sorting: O(n log n)
8	Number of LIS	DP	LC	https://leetcode.com/problems/number-of-longest-increasing-subsequence/	Track (length, count) pair at each i
9	Russian Doll Envelopes — 2D LIS	DP	LC	https://leetcode.com/problems/russian-doll-envelopes/	Sort w asc h desc, LIS on h
10	Longest Bitonic Subsequence	DP	GFG	https://practice.geeksforgeeks.org/problems/longest-bitonic-subsequence0824/	LIS from left + LDS from right -1

#	Problem	Topic	Plat	Link	Key Insight
11	Distinct Subsequences	DP	LC	https://leetcode.com/problems/distinct-subsequences/	Count ways to match T in S
12	Interleaving String	DP	LC	https://leetcode.com/problems/interleaving-string/	dp[i][j] from s1[i] or s2[j] match
13	Maximum Length of Pair Chain	Greedy/DP	LC	https://leetcode.com/problems/maximum-length-of-pair-chain/	Sort by end, activity selection
14	Longest Arithmetic Subsequence	DP	LC	https://leetcode.com/problems/longest-arithmetic-subsequence/	dp[i][diff] = dp[j][diff]+1

Total PAT-16: 14 questions

PAT-17 — Knapsack Family [12 Questions]

When to use: 0-1: each item used once. Unbounded: items reusable. Bounded: limited counts. 1D rolling array is standard.

#	Problem	Topic	Plat	Link	Key Insight
1	0-1 Knapsack — Classic	DP	GFG	https://practice.geeksforgeeks.org/problems/0-1-knapsack-problem0945/	dp[w] = max(dp[w], val+dp[w-wt])
2	Partition Equal Subset Sum	DP	LC	https://leetcode.com/problems/partition-equal-subset-sum/	Knapsack: can we reach sum/2?
3	Subset Sum — Count Ways	DP	GFG	https://practice.geeksforgeeks.org/problems/perfect-sum-problem5480/	dp[s] += dp[s-num]
4	Count Partitions with Given Difference	DP	GFG	https://practice.geeksforgeeks.org/problems/partitions-with-given-difference/	(total+diff) must be even; knapsack
5	Target Sum — Assign +/-	DP	LC	https://leetcode.com/problems/target-sum/	Count subsets with sum (total+target)/2
6	Minimum Subset Sum Difference	DP	GFG	https://practice.geeksforgeeks.org/problems/minimum-sum-partition/	Knapsack, find closest to sum/2
7	Coin Change — Unbounded	DP	LC	https://leetcode.com/problems/coin-change/	dp[amt] = min(dp[amt-c]+1), coins reusable
8	Coin Change II — Count Ways	DP	LC	https://leetcode.com/problems/coin-change-2/	dp[amt] += dp[amt-c]
9	Rod Cutting	DP	GFG	https://practice.geeksforgeeks.org/problems/rod-cutting0840/	Unbounded: dp[n] = max(price[i]+dp[n-i])
10	Ones and Zeroes — 2D Knapsack	DP	LC	https://leetcode.com/problems/ones-and-zeroes/	dp[m][n] = max with m zeros n ones
11	Last Stone Weight II	DP	LC	https://leetcode.com/problems/last-stone-weight-ii/	Knapsack: minimise difference
12	Profitable Schemes — 3D Knapsack	DP	LC	https://leetcode.com/problems/profitable-schemes/	dp[members][profit] with modular count

Total PAT-17: 12 questions

PAT-18 — Interval DP [10 Questions]

When to use: dp[i][j] = optimal answer for interval [i,j]. Fill by increasing interval length. O(n^3) typical.

#	Problem	Topic	Plat	Link	Key Insight
1	Matrix Chain Multiplication	DP	GFG	https://practice.geeksforgeeks.org/problems/matrix-chain-multiplication0309/	dp[i][j] = min cost, all split k
2	Burst Balloons	DP	LC	https://leetcode.com/problems/burst-balloons/	Last balloon k in [i,j]: left*k*right
3	Palindrome Partitioning Min Cuts	DP	LC	https://leetcode.com/problems/palindrome-partitioning-ii/	dp[i][j]=0 if palindrome else min cut
4	Minimum Score Triangulation	DP	LC	https://leetcode.com/problems/minimum-score-triangulation-of-polygon/	Triangle with each base edge
5	Minimum Cost to Merge Stones	DP	LC	https://leetcode.com/problems/minimum-cost-to-merge-stones/	Interval DP with feasibility check
6	Strange Printer — Interval DP	DP	LC	https://leetcode.com/problems/strange-printer/	Merge same-char prints
7	Scramble String — Memo Interval DP	DP	LC	https://leetcode.com/problems/scramble-string/	Split + swap check, memoised
8	Optimal Strategy Game — Coins	DP	GFG	https://practice.geeksforgeeks.org/problems/optimal-strategy-for-a-game/	Max gain for current player
9	Boolean Parenthesization — Count True	DP	GFG	https://practice.geeksforgeeks.org/problems/boolean-parenthesization/	Count T/F for each sub-expression
10	Zuma Game	DP	LC	https://leetcode.com/problems/zuma-game/	Interval DP on consecutive groups

Total PAT-18: 10 questions

PAT-19 — Heap Patterns [12 Questions]

When to use: Min/max heap for Top-K, scheduling, and merge problems. Two heaps for running median.

#	Problem	Topic	Plat	Link	Key Insight
1	Kth Largest Element in Array	Heap	LC	https://leetcode.com/problems/kth-largest-element-in-an-array/	Min heap of size K
2	Top K Frequent Elements	Heap	LC	https://leetcode.com/problems/top-k-frequent-elements/	Freq map + min heap on freq
3	K Closest Points to Origin	Heap	LC	https://leetcode.com/problems/k-closest-points-to-origin/	Max heap of size K on distance
4	Find Median from Data Stream	Heap	LC	https://leetcode.com/problems/find-median-from-data-stream/	Max heap lower + min heap upper
5	Sliding Window Median	Heap	LC	https://leetcode.com/problems/sliding-window-median/	Two heaps + lazy HashMap deletion
6	Merge K Sorted Lists	Heap	LC	https://leetcode.com/problems/merge-k-sorted-lists/	Min heap (val, listIndex)
7	Task Scheduler	Heap	LC	https://leetcode.com/problems/task-scheduler/	Max heap: schedule most frequent
8	Reorganize String	Heap	LC	https://leetcode.com/problems/reorganize-string/	Max heap: alternate most frequent
9	IPO — Maximise Capital	Heap	LC	https://leetcode.com/problems/ipo/	Sort capital, unlock into profit heap

#	Problem	Topic	Plat	Link	Key Insight
10	Smallest Range Covering K Lists	Heap	LC	https://leetcode.com/problems/smallest-range-covering-elements-from-k-lists/	Min heap + global max
11	Maximum Performance of Team	Heap	LC	https://leetcode.com/problems/maximum-performance-of-a-team/	Sort efficiency desc, heap on speed
12	Find K Pairs with Smallest Sums	Heap	LC	https://leetcode.com/problems/find-k-pairs-with-smallest-sums/	Min heap: push (i,j+1) and (i+1,0)

Total PAT-19: 12 questions

PAT-20 — Greedy Patterns [12 Questions]

When to use: Choose local optimum. Prove via exchange argument. Often combines with sort or heap.

#	Problem	Topic	Plat	Link	Key Insight
1	Activity Selection / Non-overlapping Intervals	Greedy	LC	https://leetcode.com/problems/non-overlapping-intervals/	Sort by end, skip overlapping
2	Jump Game I — Reachability	Greedy	LC	https://leetcode.com/problems/jump-game/	Track max reachable index
3	Jump Game II — Min Jumps	Greedy	LC	https://leetcode.com/problems/jump-game-ii/	BFS layers on array
4	Gas Station — Valid Start	Greedy	LC	https://leetcode.com/problems/gas-station/	Total must be >= 0; reset on fail
5	Partition Labels — Fewest Parts	Greedy	LC	https://leetcode.com/problems/partition-labels/	Last occurrence + extend boundary
6	Candy — Two-Pass	Greedy	LC	https://leetcode.com/problems/candy/	L-to-R then R-to-L, take max
7	Minimum Number of Arrows	Greedy	LC	https://leetcode.com/problems/minimum-number-of-arrows-to-burst-balloons/	Sort by right end
8	Fractional Knapsack	Greedy	GFG	https://practice.geeksforgeeks.org/problems/fractional-knapsack-1587115620/	Sort by value/weight ratio
9	Job Sequencing — Max Profit	Greedy	GFG	https://practice.geeksforgeeks.org/problems/job-sequencing-problem-1587115620/	Sort by profit desc, fill slots
10	Queue Reconstruction by Height	Greedy	LC	https://leetcode.com/problems/queue-reconstruction-by-height/	Sort h desc k asc, insert at k
11	Two City Scheduling	Greedy	LC	https://leetcode.com/problems/two-city-scheduling/	Sort by (costA-costB), first half to A
12	Maximum Profit in Job Scheduling	Greedy+DP	LC	https://leetcode.com/problems/maximum-profit-in-job-scheduling/	Sort by end, DP+BS last non-overlap

Total PAT-20: 12 questions

PAT-21 — Backtracking Patterns [12 Questions]

When to use: Try all options at each step, recurse, undo (backtrack). Prune invalid branches early.

#	Problem	Topic	Plat	Link	Key Insight
1	Subsets — All 2^n	Backtracking	LC	https://leetcode.com/problems/subsets/	Include/exclude at each index
2	Subsets II — With Duplicates	Backtracking	LC	https://leetcode.com/problems/subsets-ii/	Sort + skip same at same depth level

#	Problem	Topic	Pl at	Link	Key Insight
3	Permutations	Backtracking	LC	https://leetcode.com/problems/permutations/	Swap-based or used[] array
4	Permutations II — Duplicates	Backtracking	LC	https://leetcode.com/problems/permutations-ii/	Sort + skip same at same level
5	Combination Sum I — Reuse	Backtracking	LC	https://leetcode.com/problems/combination-sum/	Recurse same index for reuse
6	Combination Sum II — No Reuse	Backtracking	LC	https://leetcode.com/problems/combination-sum-ii/	Skip same element at same depth
7	Palindrome Partitioning	Backtracking	LC	https://leetcode.com/problems/palindrome-partitioning/	Try every palindrome prefix, recurse
8	N-Queens	Backtracking	LC	https://leetcode.com/problems/n-queens/	Row by row, 3 boolean arrays
9	Sudoku Solver	Backtracking	LC	https://leetcode.com/problems/sudoku-solver/	Try 1-9, backtrack if invalid
10	Word Break II — All Sentences	Backtracking	LC	https://leetcode.com/problems/word-break-ii/	If prefix in dict, recurse on rest
11	Expression Add Operators	Backtracking	LC	https://leetcode.com/problems/expression-add-operators/	Track eval and last-multiplied
12	Path with Maximum Gold	Backtracking	LC	https://leetcode.com/problems/path-with-maximum-gold/	DFS all 4 dirs, mark 0, restore

Total PAT-21: 12 questions

PAT-22 — Trie Patterns [10 Questions]

When to use: O(L) prefix lookup. Insert words and traverse to answer prefix/autocomplete queries.

#	Problem	Topic	Pl at	Link	Key Insight
1	Implement Trie	Trie	LC	https://leetcode.com/problems/implement-trie-prefix-tree/	TrieNode[26] + isEnd flag
2	Add and Search with Wildcards	Trie	LC	https://leetcode.com/problems/design-add-and-search-words-data-structure/	DFS on '.' branches all children
3	Replace Words with Root	Trie	LC	https://leetcode.com/problems/replace-words/	Trie: find shortest prefix match
4	Maximum XOR of Two Numbers	Trie	LC	https://leetcode.com/problems/maximum-xor-of-two-numbers-in-an-array/	Binary trie, greedy opposite bit
5	Maximum XOR Subarray	Trie	GFG	https://practice.geeksforgeeks.org/problems/max-xor-subarray1825/	Prefix XOR + trie
6	Word Search II — Multiple Words	Trie	LC	https://leetcode.com/problems/word-search-ii/	Trie of words + DFS on grid
7	Design Autocomplete System	Trie	LC	https://leetcode.com/problems/design-search-autocomplete-system/	Trie + DFS collect top-3 by freq
8	Palindrome Pairs	Trie	LC	https://leetcode.com/problems/palindrome-pairs/	Trie of reversed words + checks
9	Concatenated Words	Trie	LC	https://leetcode.com/problems/concatenated-words/	Trie + DFS: word = 2+ shorter words
10	Count Distinct Substrings	Trie	GFG	https://practice.geeksforgeeks.org/problems/count-of-distinct-substrings/	Insert all suffixes, count new nodes

Total PAT-22: 10 questions

PAT-23 — Union-Find Patterns [10 Questions]

When to use: Dynamic connectivity. Union by rank + path compression = nearly O(1) per operation.

#	Problem	Topic	Plat	Link	Key Insight
1	Number of Provinces	UF	LC	https://leetcode.com/problems/number-of-provinces/	Union connected cities, count roots
2	Redundant Connection	UF	LC	https://leetcode.com/problems/redundant-connection/	Edge whose both nodes share root
3	Accounts Merge	UF	LC	https://leetcode.com/problems/accounts-merge/	Union emails, collect by root
4	Satisfiability of Equality Equations	UF	LC	https://leetcode.com/problems/satisfiability-of-equality-equations/	Union == then check != for conflict
5	Min Cost to Connect All Points — Kruskal	UF	LC	https://leetcode.com/problems/min-cost-to-connect-all-points/	Sort O(n^2) edges, Kruskal+UF
6	Remove Max Edges Keep Graph Traversable	UF	LC	https://leetcode.com/problems/remove-max-number-of-edges-to-keep-graph-fully-traversable/	Two UF sets, prefer type-3 first
7	Regions Cut by Slashes	UF	LC	https://leetcode.com/problems/regions-cut-by-slashes/	Each cell = 4 triangles, union by slash
8	Swim in Rising Water	UF	LC	https://leetcode.com/problems/swim-in-rising-water/	Sort cells by height, union as raised
9	Making a Large Island	UF	LC	https://leetcode.com/problems/making-a-large-island/	Color components, flip each 0 greedily
10	Number of Islands II — Online	UF	LC	https://leetcode.com/problems/number-of-islands-ii/	Dynamic union with each land add

Total PAT-23: 10 questions

PAT-24 — Bit Manipulation [10 Questions]

When to use: XOR for pairing, bit masks for state, n & (n-1) to clear lowest set bit. O(1) tricks for math.

#	Problem	Topic	Plat	Link	Key Insight
1	Single Number — XOR	Bit	LC	https://leetcode.com/problems/single-number/	XOR all: pairs cancel, unique remains
2	Single Number II — Appears 3 Times	Bit	LC	https://leetcode.com/problems/single-number-ii/	Bit vote: count mod 3 per bit
3	Single Number III — Two Unique	Bit	LC	https://leetcode.com/problems/single-number-iii/	XOR all, split by any set bit
4	Missing Number — XOR	Bit	LC	https://leetcode.com/problems/missing-number/	XOR indices 0..n with array values
5	Number of 1 Bits	Bit	LC	https://leetcode.com/problems/number-of-1-bits/	n & (n-1) clears lowest set bit
6	Counting Bits — All 0..n	Bit	LC	https://leetcode.com/problems/counting-bits/	dp[i] = dp[i >> 1] + (i & 1)
7	Power of Two	Bit	LC	https://leetcode.com/problems/power-of-two/	n > 0 && (n & (n-1)) == 0
8	Sum Without + Operator	Bit	LC	https://leetcode.com/problems/sum-of-two-integers/	XOR = sum, AND << 1 = carry

#	Problem	Topic	Plat	Link	Key Insight
9	Subsets via Bitmask	Bit	LC	https://leetcode.com/problems/subsets/	Each bit = include/exclude element
10	Maximum AND Sum of Array	Bit+DP	LC	https://leetcode.com/problems/maximum-and-sum-of-array/	Bitmask DP on slot assignments

Total PAT-24: 10 questions

Pattern Summary — P25

Pattern	Name	Count
PAT-01	Two Pointers	14
PAT-02	Sliding Window	14
PAT-03	Binary Search Classic	14
PAT-04	Binary Search on Answer	12
PAT-05	Prefix Sum + HashMap	12
PAT-06	Monotonic Stack	12
PAT-07	Tree — Traversal and Views	14
PAT-08	Tree — Advanced (Paths, LCA, Construction)	14
PAT-09	BST Patterns	12
PAT-10	Graph — BFS	14
PAT-11	Graph — DFS	12
PAT-12	Shortest Path	12
PAT-13	Topological Sort	10
PAT-14	1D Dynamic Programming	18
PAT-15	2D Grid DP	12
PAT-16	LCS / LIS Family	14
PAT-17	Knapsack Family	12
PAT-18	Interval DP	10
PAT-19	Heap Patterns	12
PAT-20	Greedy Patterns	12
PAT-21	Backtracking Patterns	12
PAT-22	Trie Patterns	10
PAT-23	Union-Find Patterns	10
PAT-24	Bit Manipulation	10
TOTAL		**298**